# Lishu Zhang

Tilburg University & CREST

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## Employment

2025–present Postdoctoral Researcher, CREST, ENSAE Paris, Institut Polytechnique de Paris

## Education

2021–present Ph.D. in Economics, Tilburg University
PhD expected: December 2026

2019–2021 Research Master in Economics, Tilburg University, with distinction

2017–2019 Master of Science in Economics, Stockholm School of Economics

2013–2017 Bachelor of Economics, Renmin University of China

## Research Interests

Financial Economics, Environmental Economics, International Economics
**Focus:** Climate Finance, Cross-border Capital Flows, Production Networks, Critical Raw Materials

## Working Papers

*Measuring Critical Raw Material Dependence (w/ Peter Tankov and Olivier David Zerbib)* ***(Job Market Paper)***

Geopolitical tensions and economic transitions make critical raw material supply disruptions a growing concern, but assessing their consequences requires knowing which firms depend on which materials. With language-model assistance, we reconstruct production paths from technical evidence and multiply input cost shares along each path to obtain a product's marginal-cost elasticity to material prices. In our model, this elasticity also governs exposure to quantity shocks. We obtain 1,760 material–product dependence estimates across 47 critical raw materials and construct firm-level measures for 81,980 firm-years covering 13,301 firms. Comparisons with specialist reports and corporate filings support the measure and reveal how selectively dependence enters public information. The network contains an exact product or clear category counterpart for 84.2% of relationships identified in 124 specialist reports, but those reports cover only 1.2% of material–product relationships in the network for overlapping materials. Measured dependence is positive for 52.9% of current operational relationships disclosed in 10-Ks, whereas filings affirm only 8.5% of positive measured dependencies. Public information thus captures a narrow, salient subset of production dependence. The measures reveal within-firm co-dependence and substantial dispersion among industry peers. Among non-resource industry peers, greater CRM dependence is also associated with higher market-to-book ratios. As an application, we study cross-sectional market responses to material supply shocks and find that investors rarely price dependencies outside public attention.

*Resistance and Arbitrage: International Trade in Brown Loans*

Yihong Xia Best Paper Award, CICF 2025.

I develop a novel measure of carbon sensitivity in lending to assess reductions in portfolio exposure to brown

assets. Using syndicated loan data, I show that countries with greater resistance to brown lending, proxied by economic development, experience faster shifts in the sectoral composition of loan portfolios. The decarbonization is driven primarily by domestic credit reallocation. I find consistent evidence of risk transfers to less regulated lenders and foreign countries, indicating arbitrage and incomplete regulations. Furthermore, lenders' climate risk-taking and transfer behaviors vary sharply by syndicate role, loan type, and specialization. The existence of international trade in brown loans has important implications for supervisory evaluation. Using the European Central Bank's climate guide, I show that accounting for regulatory leakage reveals effects contrary to common wisdom.

*Who holds brown equity? Economic development, divestment, and environmental outcomes (w/ Harald Benink, Harry Huizinga, and Louis Raes)* (Under review)

Using global equity-ownership data for 2002–2021 covering 3,559 firms and 111 investor countries, we show that investors in more developed countries hold less carbon-intensive portfolios. This income gradient appears within varying investor types and remains after accounting for return chasing and financial development, consistent with environmental concerns being more strongly reflected in richer-country portfolios. Greater ownership by poorer-country investors predicts higher subsequent emissions and ESG incidents. Exploiting waves of fossil-fuel divestment commitments, we construct a shift-share measure using firms' predetermined ownership. Rich-country divestment pressure shifts ownership toward poorer countries and worsens environmental performance, reduced-form effects consistent with an ownership channel.

*Does ESG information shape control? Evidence from takeovers*

When two firms combine into one, the allocation of control determines who manages their assets. We study whether ESG information shapes control using 7,914 completed deals announced from 2003 to 2025. We estimate that ESG rating availability increases the rated firm's predicted probability of taking control by 35–37 percentage points between two otherwise equal firms. Where both firms are rated, control tends to go to the higher-scoring firm, most strongly on the Environmental pillar across borders. The information advantage is larger under strong audit standards and smaller where corporate reporting is widespread. Deal-level announcement returns do not differ significantly across combinations of acquirer and target rating status, but after 2016 a rated target bought by an unrated firm earns less than an unrated target bought by a rated firm.

## Work in Progress

*Localized Impacts of Transition Risks (w/ Thibaud Barreau, Jakob Kellermann, Peter Tankov, and Olivier David Zerbib)*

We develop a facility-level measure of transition risk based on the profit losses that would arise if carbon prices moved toward local social-cost-of-carbon benchmarks. The framework combines plant emissions, effective policy coverage, abatement options, and uncertain repricing dates. By matching facilities to detailed carbon-pricing rules, it captures variation that national headline prices and firm-level emissions measures miss. The next stage will connect this exposure map to the local economic and environmental effects of realized transition shocks.

## Teaching Experience

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| Spring 2022–2024 | Teaching Assistant, *Macroeconomics 3*, PhD level, Tilburg University |
| Fall 2021–2024 | Teaching Assistant, *Economics of the European Union*, undergraduate level, Tilburg University |
| Fall 2021–2022 | Teaching Assistant, *Economics of Banking and Finance*, undergraduate level, Tilburg University |
| Fall 2018 | Teaching Assistant, *International Economics*, undergraduate level, Stockholm School of Economics |

## Grants, Scholarships, and Awards

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| 2026 | AFA Travel Grant, American Finance Association |
| 2025 | Yihong Xia Best Paper Award, CICF |
| 2019–2021 | Koopmans Scholarship, Tilburg University |
| 2015 | National Scholarship, Renmin University of China |
| 2014 | Outstanding School Cadre, Renmin University of China |

## Seminar & Conference Presentations

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| 2026 | AFA Poster Session; Junior Finance Research Day, CREST; CEMRACS; HEC Paris Finance PhD Workshop |
| 2025 | CICF; AsianFA PhD Consortium; PhD-Economics Virtual Seminar |
| 2023–2024 | FMA; Canadian Economics Association; DEARE, Utrecht University; ESSFM, Gerzensee; Imperial College PhD Conference; KU Leuven; Tilburg University (Finance, Economics) |

## University Service

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| 2021–2023 | Member, Tilburg PhD Platform (TiPP) |
| 2019–2020 | Representative of Research Master's in Economics, Graduate Students' Society (GSS) |

## Computer Skills

STATA, R, MATLAB, Python, LaTeX.

## Languages

English, French (intermediate), Mandarin

## References

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